

Serie 12

RANK AND INVERTIBILITY

1. Rank. Suppose that V, W are finite-dimensional vector spaces over a field K . Let $T \in \text{Hom}(V, W)$. Prove that $\text{rank}(T) = 1$ if and only if there exists a basis $\mathcal{B}$ of V and a basis $\mathcal{C}$ of W such that with respect to these bases, all entries of $[T]_{\mathcal{C}}^{\mathcal{B}}$ equal 1.

2. Invertible matrices. Suppose V is finite-dimensional and $S, T, U \in \text{Hom}(V, V)$ with $STU = \text{Id}_V$. Show that T is invertible and that $T^{-1} = US$.

3. Invertible matrices. Let V be a finite-dimensional vector spaces over a field K . Suppose that $T \in \text{Hom}(V, V)$, and that $\mathcal{A} = \{u_1, \dots, u_n\}$ and $\mathcal{B} = \{v_1, \dots, v_n\}$ are bases of V . Prove that the following are equivalent:

- (a) T is invertible.
- (b) The columns of $[T]_{\mathcal{B}}^{\mathcal{A}}$ are linearly independent in K_{col}^n ;
- (c) The columns of $[T]_{\mathcal{B}}^{\mathcal{A}}$ span K_{col}^n ;
- (d) The rows of $[T]_{\mathcal{B}}^{\mathcal{A}}$ are linearly independent in K_{row}^n ;
- (e) The rows of $[T]_{\mathcal{B}}^{\mathcal{A}}$ span K_{row}^n .

4. Morphisms of vector spaces. Let V be a finite-dimensional vector space. Prove or disprove:

- (a) Let $V' \subset V$ be a linear subspace. Every automorphism $f : V' \rightarrow V'$ can be extended to an automorphism $\bar{f} : V \rightarrow V$.
- (b) For every Endomorphism $f : V \rightarrow V$ its image $\text{Im}(f)$ is a linear complement $\text{Ker}(f)$ in V .
- (c) There does not exist any linear map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^5$ such that

$$\text{rank}(T) = \dim \text{Ker}(T).$$

5. Powers of matrices. Consider the space $M_{2 \times 2}(K)$ of 2-by-2 matrices over a field K .

- (a) Show that if $A \in M_{2 \times 2}(K)$ satisfies $A^2 \neq 0$, then $A^k \neq 0$ for all $k \geq 3$.
- (b) Find a field K and a matrix $A \in M_{2 \times 2}(K) \setminus \{0\}$ such that $\exists n \in \mathbb{N} : A^n = 0$.

6. Rank of matrices. Determine the ranks of the following rational $n \times n$ -matrixes, those being elements of $M_{n \times n}(\mathbb{Q})$, depending on the positive integer n .

- (a) $(kl)_{k,l=1,\dots,n}$;
- (b) $((-1)^{k+l}(k+l-1))_{k,l=1,\dots,n}$;
- (c) $\left(\frac{(k+l)!}{k!l!} \right)_{k,l=0,\dots,n-1}$.

Hint: Note that the last matrix is indexed from 0 to $n-1$. You may use induction to find the desired formula.